

Methods: J0604

ALMA Thesis Planner (automated pipeline)

0.1 0. Data provenance, retrieval, and the coverage census (Week 0–1)

Everything in this thesis comes from one place: the delivered products of ALMA project 2023.1.00804.S. Start by querying the ALMA Science Archive on that project code alone and pulling the full delivery for every target in the program — raw ASDMs plus the calibration and imaging scripts, the pipeline weblogs, and whatever calibrated measurement sets and pipeline continuum images and cubes are served. Restore the calibration with the delivered ‘scriptForPI.py’ / pipeline restore task under a single pinned CASA version, and record that version number; do not mix CASA versions across the sample, because the frequency-frame machinery and ‘tclean’ gridder defaults have changed between releases and you are about to make a measurement whose entire value lies in frequency precision.

Immediately build two tables and keep them under version control for the rest of the year.

Table A — delivery inventory. One row per target per spectral window, populated by reading the ‘SPECTRAL_WINDOW’ sub-table of each measurement set (‘msmd’, ‘listobs’, or ‘casatools.table’ directly) and the FITS headers of the delivered cubes: spw ID, number of channels, channel width, total bandwidth, the reference frequency and the *frame* in which it is quoted (the MS carries TOPO; the pipeline cubes may be LSRK — never assume, always read it), the on-source integration time, the achieved continuum rms and beam from the weblog, and the number of antennas and the array configuration. Also record the adopted tuning redshift the proposal team used for each target, taken from the delivery itself (the source table / spw placement). Treat that value as a bookkeeping number with no error bar: it exists in this thesis for exactly one purpose, which is to tell you where in the band to look.

Table B — rest frequencies. Pull the ground-state transitions of H_2O^+ , and of OH^+ , from a standard laboratory line list (CDMS and/or JPL, accessed through Splatalogue). Record for each transition the exact tabulated frequency, the quantum numbers, the tabulated uncertainty, the catalog, and the date and version you retrieved it. These are the only non-project numbers in the thesis and they are physical constants, so document them as such. The ground states of these hydrides are split into fine- and hyperfine-structure components that are blended at the line widths we expect in starbursts, so you must adopt a single explicit convention: define the reference rest frequency of each transition as the intensity-weighted mean of its tabulated hyperfine components, compute it yourself from the catalog line strengths, and use the *same* convention for every transition. A differential velocity measurement is only as good as the consistency of this convention, so write the calculation into a script rather than doing it by hand.

The coverage matrix. Now cross Table A with Table B. For each target, redshift each tabulated rest frequency by that target’s tuning redshift and ask whether it lands inside a delivered spw. Require a margin: the predicted line center must sit at least half of your widest search kernel

away from the spw edge, otherwise flag it "edge" rather than "covered," because a line whose wing runs off the band cannot have its centroid or equivalent width measured. The output is a $\text{targets} \times \text{transitions}$ matrix with entries {covered, edge, not covered}. This matrix — not the proposal abstract — defines your working sample, and in particular it tells you empirically which pairs of transitions (the two fine-structure components of the H_2O^+ ground state, and any OH^+ ground-state transition that happens to fall in a delivered window) are available for the differential measurement in step 5. Do this in Week 1 and present it as your first thesis figure, because the size of the two-transition subsample determines whether you are executing the main plan or the contingency.

If part of the delivery is still proprietary. Take the sample that is genuinely in hand, and in parallel build and validate steps 2–4 end to end on synthetic spectra generated from the delivery's own header parameters — the real channelizations, the real spw layouts, the real measured rms values from Table A — with known injected lines. When the remaining data release, you run the frozen pipeline on it. Do not wait for data to start writing code.

0.2 1. Uniform continuum photometry (Weeks 1–5)

The continuum is the backlight for every absorption measurement, so it is measured first and measured identically for everyone.

Image the aggregate continuum for each target with 'tclean' in 'mfs' mode, using the line-free channels of all spws (defined by the iterative masking in step 2 and re-run once that masking is final), with a fixed imaging prescription applied sample-wide: Briggs weighting at a single robust value, 'auto-multithresh' masking with fixed parameters, cleaning to a fixed multiple of the per-image rms, and 'pbcor=True' output retained alongside the flat-noise image. Record the restoring beam and the rms measured in a signal-free annulus of the flat-noise image.

Photometry uses a curve of growth: place concentric circular apertures of increasing radius on the primary-beam-corrected image, centered on the flux-weighted centroid, and adopt the radius at which the enclosed flux reaches a stated fraction of its asymptote, with that fraction fixed for the whole sample and stated in the thesis. For sources consistent with unresolved, fall back to a 2D Gaussian fit with 'imfit'; classify a source as resolved only when the deconvolved major axis exceeds its own uncertainty by a stated significance and when the integrated-to-peak flux ratio is inconsistent with unity. Record the resolved/unresolved classification per source — it feeds the spot-checks in step 4.

Uncertainties come from the data, not from the fit: bootstrap by dropping many apertures of the adopted size at random signal-free positions across the primary-beam-corrected residual image (weighting by the primary-beam response so you are not comparing apertures at different sensitivities), and take the scatter of the recovered sums. Add the ALMA flux-scale systematic from the weblog in quadrature only where an absolute flux enters; note in the text that it does *not* enter the continuum-normalized quantities that follow. Deliverable: the sample's continuum brightness distribution and the resolved/unresolved classification, which is the denominator of every optical depth you will quote.

0.3 2. Uniform spectral extraction (Weeks 5–11)

Image every delivered spw of every target as a cube with the same weighting scheme as the continuum, on a common velocity channel width chosen as the coarsest native resolution in the sample so

that no source is artificially upweighted; rebin with a flux-conserving scheme and re-measure the noise empirically after rebinning rather than propagating it analytically, because ALMA channels are correlated.

Continuum subtraction is a fork in the road, so settle it once, early, and in writing. Test uv-plane subtraction ('uvcontsub', low-order polynomial fit over line-free channels) against image-plane subtraction ('imcontsub' or a per-pixel polynomial on the cube) on two representative sources spanning the brightness range — one bright and resolved, one faint and unresolved. Define line-free channels iteratively: fit, search, mask any candidate feature, refit. Compare the resulting normalized spectra channel by channel, fix one method sample-wide, and carry the median absolute difference between the two methods as a documented systematic that you propagate into the equivalent widths and centroids in step 5.

Extract spectra through the *same* aperture prescription established in step 1, then divide by the continuum measured in that identical aperture, giving a continuum-normalized spectrum per source per spw. Convert to apparent optical depth as $\tau_{\text{app}}(\nu) = -\ln[S(\nu)/S_{\text{cont}}]$, and state explicitly in your methods chapter that this is an apparent optical depth defined with a covering fraction of unity — it is a definition of the reported quantity, not an assumption you will later relax.

Regrid all cubes to a single, explicitly named frequency frame before any centroid is measured, and apply the identical 'mstransform'/'cvel2' conversion to every spw of a given target. This matters: the differential measurement in step 5 needs no systemic redshift, but it *does* need both transitions of a given source to live on the same frequency scale.

0.4 3. Matched-filter feature search (Weeks 11–17)

Build a filter bank of Gaussian kernels logarithmically spaced in FWHM from roughly 50 to 800 km s⁻¹, spanning the range of widths plausible for these systems, and cross-correlate each normalized spectrum with every kernel. Normalize each filter so that its output on pure noise has unit variance, and calibrate that normalization empirically per spectrum rather than trusting the nominal rms. Search both signs of the response — absorption and emission — over every covered transition window and, for threshold calibration, over the full line-free extent of every spw.

Order matters here. These are bright starbursts and the sidebands contain real emission lines; if you calibrate your false-alarm rate on spectra that still contain them, you inflate the noise estimate and silently destroy your absorption sensitivity. So: run a first pass, mask every candidate emission feature above a permissive threshold, recompute the noise, then run the real search. Calibrate the false-alarm rate empirically by running the identical filter bank on off-source apertures of the same size in the same cubes and on the masked line-free regions, building the distribution of maximum filter response under the null, and setting the detection threshold so that the expected number of false positives across the entire sample-wide trials factor (sources × transitions × filters × independent channels) is below a stated small number. Quote that trials factor.

Fit every surviving candidate in the observed-frequency domain with a Gaussian (and, where residuals are structured, a Gauss–Hermite profile), using least squares for the initial solution and an MCMC sampler for the posteriors, so that centroid, width, depth, and their covariances come out together. Report per source and per transition: peak apparent optical depth, equivalent width, FWHM, and observed-frequency centroid ν_{obs} , each with uncertainties. Inspect every fit by eye against the spectrum and the residual — no candidate enters the catalog unlooked-at.

0.5 4. Injection–recovery in the image plane (Weeks 17–24)

Completeness is measured, not assumed. Inject synthetic Gaussian absorption profiles directly into the real continuum-normalized spectra at random frequencies that avoid the real features you just catalogued, on a grid of peak optical depth $\times$ FWHM, with many independent trials per grid cell. Re-run the *unmodified* search and fit pipeline blind — the recovery code must not know where the injections went. Output per source: a completeness surface over $(\tau_{\text{peak}}, \text{FWHM})$, from which you quote τ_{90} , the peak optical depth recoverable at 90% completeness for a fiducial 200 km s^{-1} feature. Record two more things from the same runs, because step 5 needs them: the bias and scatter of the recovered *centroid* and the recovered *width* as a function of S/N.

Visibility-plane injection is deliberately not done — it is an unbounded CASA compute burden on a 40-source sample. Instead, bound the imaging-stage flux losses analytically from the array configuration, the source sizes from step 1, and the cleaning depth, and spot-check that bound on the two or three sources classified as resolved by comparing an image-plane measurement against a direct uv-domain fit of the same feature. State the bound and the spot-check results in the text.

0.6 5. The differential measurement (Weeks 24–33)

This is the science, and it is built so that no systemic redshift ever enters.

For each source in which two hydride transitions are both covered and both measured, convert each observed centroid to its own independent redshift, $z_i = \nu_{\text{rest},i} / \nu_{\text{obs},i} - 1$, using the hyperfine-weighted rest frequencies from Table B, and form the velocity difference $\Delta v = c (z_1 - z_2) / (1 + \bar{z})$. Say this plainly in the thesis: each transition supplies its own redshift from its own sky frequency, so the difference of the two contains no adopted systemic redshift at all, and any error in the tuning redshift used to *locate* the windows cannot propagate into Δv . Propagate the two centroid uncertainties from the MCMC posteriors, add the continuum-subtraction systematic from step 2 and the centroid bias from step 4, and treat the frequency-scale term as common-mode when both lines lie in the same spw and as a residual additive term when they do not.

Report alongside Δv two dimensionless ratios: the ratio of equivalent widths and the ratio of fitted FWHMs, both computed as log-ratios so their uncertainties are symmetric, with posteriors drawn jointly from the two fits.

Then test the ensemble. Form the inverse-variance-weighted mean Δv over the two-transition subsample and test it against a permutation null: randomly permute the transition labels (equivalently, randomize the sign of each Δv) many thousands of times, rebuild the weighted mean each time, and read the p-value off that distribution. Cross-check the uncertainty on the mean with a bootstrap over sources. A systematic nonzero mean offset means the two ions occupy kinematically distinct layers; a null result is reported as an upper limit on any mean offset at a stated precision in km s^{-1} , and that number is the headline either way — so quote it prominently.

For sources too faint to yield individual centroids, stack. Align spectra on the adopted tuning redshift, resample to a common velocity grid, and stack each transition *separately*, forming both an inverse-variance-weighted mean and a median stack, with jackknife-over-sources uncertainties. The logic that makes this work is worth stating explicitly: an error in the alignment redshift displaces both transitions of a given source by the same velocity, so both stacks are smeared by the identical kernel and the *difference* of the two stack centroids survives the smearing. Quantify the residual bias in that difference by pushing the step-4 injected profiles through the identical stacking machinery and measuring what comes out.

0.7 6. Sensitivity-matched census, deliverables, and write-up (Weeks 33–38)

For every source and every covered transition, publish either a detection with its fitted parameters or a completeness-calibrated upper limit — use τ_{90} from step 4, not a naive rms-scaled limit, since the latter would misstate what the search could actually have found. Compute the incidence rate only within the subsample whose completeness surfaces all reach a common fiducial τ threshold, so that shallow and deep pointings are never averaged into the same number, and give binomial confidence intervals (Wilson or Jeffreys) on that rate. Then test whether incidence still correlates with continuum brightness *at matched sensitivity*, again with a permutation test: shuffle the detection flags among the sensitivity-matched sources many times and compare the observed brightness split against the null distribution.

Fix the reported quantities now and do not drift: apparent peak optical depth, equivalent width, FWHM, observed-frequency centroid, velocity differences, and the two dimensionless ratios. Column densities, abundances, fractional ionization, and mass or momentum flux rates are not computed here, because they require a covering fraction, a geometry, and an excitation model that this measurement set does not constrain; OH^+ appears as a stated sample-selection property with no numbers attached unless one of its transitions is found in this project’s own spectral windows by the Week-1 coverage matrix.

Deliver three artifacts with the thesis: the machine-readable feature catalog (one row per source per transition, detections and limits together, with all uncertainties and flags), the per-source completeness surfaces in a documented FITS or HDF5 format, and the analysis scripts in a git repository with the CASA version, the Python environment, and the Table B catalog retrieval date pinned in the README. Every figure must be regenerable from the repository by a single command. Carry the required ALMA acknowledgement for ADS/JAO.ALMA#2023.1.00804.S and the standard NRAO acknowledgement verbatim in the thesis and in any resulting paper.

If the Week-1 coverage matrix shows that a second transition is covered in too few sources to support step 5, you run steps 1–4 and 6 in full and the thesis is the sensitivity-matched incidence census with completeness-calibrated optical-depth limits for a 40-source absorption survey — build the pipeline so that this branch requires deleting nothing, only stopping earlier.